

GRANT BENSON

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Experience

Prometheus Group

September 2025 - Current

SDET | C#, Playwright, Jira

- Architected the Playwright/C# automation pipeline from scratch for MDaaS, establishing the core test strategy and release schedule for all development and release candidate builds.
- Enforced CI/CD pipeline stability in GitLab by integrating the Xray test suite for automated reporting, auditing daily test runs, and resolving regressions to minimize release blockers.
- Validated complex end-to-end EAM/ERP integration pipelines across SAP and IBM Maximo, ensuring reliable data synchronization for enterprise workflows.
- Spearheaded cross-team automation initiatives: built a k6 load suite scaling to 160 virtual users for STO and automated critical Vulcan modal workflows.

North Carolina State University

February 2025 – May 2025

College of Engineering Undergraduate Researcher | Python, mmWave Studio

- Built a system to generate animated 3D skeletons from synchronized radar point cloud and multi-IMU data.
- Developed Python tools for depth estimation, sensor fusion, and real time human motion visualization.

The Net VR

May 2024 – November 2024

Gameplay Programmer | C#, Unity

- Engineered alpha and beta multiplayer builds, managing feature delivery and product iterations across Scrum sprints.
- Implemented core gameplay systems including client input, animation, Vivox spatial sound, and an object manager to create immersive digital cyberspace environment.

Projects

FoldeD 2.0 | C#, Unity

April 2026 – Current

- Implemented CVT-MAP-Elites to search paper airplane fold geometries, tessellating a behavior space too high-dimensional for grid archives, surfacing structurally distinct high-glide designs over a single optimum.
- Designed a variable-length genome encoding for fold sequences with crossover and mutation operators that preserve fold validity, scoring candidates through automated simulation rounds to facilitate unsupervised learning.

LAIGO Mosaic Maker | Python, FastAPI, React 19, Vite, Render, Stripe

November 2025 – July 2026

- Chained image segmentation, color quantization, and mosaic layout synthesis AI models into an async pipeline to procedurally generate purchasable instructions and LEGO piece lists with real-time job queue polling.

Anti Angler | Python, PyTorch, YOLOv8, FastAPI, rasterio/GeoPandas

August 2026

- Trained vessel detectors to boost held-out recall from 0.08 to 0.77 (0.94 precision on dense anchorages) using a FastAPI review tool to drive an active-learning loop, fusing Sentinel-1/2, AIS, and Global Fishing Watch data.
- Built an automated evaluation pipeline sweeping 1,799 marine areas nightly in CI backed by 355 offline tests.

ClaimAR | TypeScript, Lens Studio

July 2025 – June 2026

- Engineered a real-time multiplayer AR experience, owning the end-to-end architecture to sync peer-replicated grid states over an eventually consistent cloud store.
- Developed exterior flood-fill algorithms for territory resolution and implemented chunked write pacing to adhere to cloud sync throughput limits.

Coffee Maker | HTML, CSS, Javascript, Java, Spring, Hibernate, MySQL

April 2024

- Built full-stack e-commerce features for order processing, fulfillment, user auth, and real time inventory management.
- Engineered automated backend test suites using JUnit and integrated GitHub Actions for CI/CD regression testing.

Education

North Carolina State University

Aug 2021 – May 2025

B.S. Computer Science & Game Development Concentration (3.5 GPA)

Raleigh, NC

- **Relevant Coursework:** Software Engineering, Artificial Intelligence, Linear Algebra, Data Structures & Algorithms, Operating Systems, Game Engines, Graphics

Technical Skills

Languages: C#, Python, C++, C, Java, JavaScript, TypeScript, SQL

Developer Tools & Technologies: Git, .NET, Playwright, React, FastAPI, PyTorch, YOLOv8, Qiskit, NumPy, k6, Postman, Three.js, WebGL, Tailwind, Lens Studio, Unity, Unreal Engine